

Table 19: Answer table for Q.22 over Table III of Fischer et al. [20], modeled in the ORKG as Comparison R1563034. The table shows, for each material class, representative materials, dominant reactant tuples (Reactant 1–3), and the number of reported ALE processes using those tuples.

Material class / representative	Dominant reactant tuples	# Reports
Oxides (Al_2O_3 , HfO_2 , ZrO_2)	HF + (TMA, DMAC, $\text{Sn}(\text{acac})_2$, TiCl_4)	25
Fluorides / Nitrides (AlF_3 , AlN)	HF + $\text{Sn}(\text{acac})_2$	2
III–V semiconductors (Ga_2O_3 , GaN, InGaAs)	HF/ BCl_3 or XeF_2 cycles with DMAC/TMA	4
Transition metals (Co, Ni, Fe, Cu)	Oxidation/chelation: O_2/O_3 or Cl_2 + Hhfac/Hacac	4
Si-based systems (Si, SiO_2 , Si_3N_4)	HF/TMA, Cl_2 , or O_2/O_3 plasma combinations	6
Refractory / conductive films (TiN, W, WO_3 , TiO_2)	$\text{WF}_6/\text{BCl}_3/\text{O}_2/\text{O}_3$ ligand-exchange routes	5
Other / emerging (VO_2 , ZnO)	HF/ SF_4 or Hacac + O_2 plasma	3

Easy Query. The 22nd question is: “List, for each material etched, the distinct reactant tuples reported (Reactant 1–3) and how many entries report each tuple.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t3-easy-ale>. The tabulated, synthesized answer is shown in Table 19. It lists material classes with their dominant reactant tuples and the number of reported ALE processes for each. In this dataset, the query returns 46 unique material–reactant tuples.

Table 20: Answer table for Q.23 over Table III of Fischer et al. [20], modeled in the ORKG as Comparison R1563034. For each mechanistic archetype, the table reports the number of distinct ALE chemistries and the number of etched materials.

Mechanism archetype	# Chemistries	# Materials
Fluorination + ligand-exchange	23	17
Other / unclassified	11	10
Oxidation + chelation	3	4
Halogenation & conversion	1	1

Complex Query. The 23rd question is: “Classify each ALE chemistry by its dominant mechanistic archetype—(i) Fluorination + ligand-exchange (HF with TMA/DMAC/ $\text{Sn}(\text{acac})_2$), (ii) Oxidation + chelation (O_2/O_3 with Hacac/Hhfac), (iii) Halogenation & conversion (Cl_2/XeF_2 with $\text{BCl}_3/\text{WF}_6/\text{WCl}_6/\text{TiCl}_4$), or (iv) Other.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t3-mechanism-buckets>. As explained earlier, answers in this work are tabulated, synthesized results based on the original question; Table 20 lists, for each mechanistic archetype, the number of distinct chemistries and etched materials in the ORKG comparison.

3.6 Paper – Thermal atomic layer etching: Mechanism, materials and prospects

Fang *et al.* (2018) [19] present a widely cited review on *thermal atomic layer etching* (ALE), consolidating chemical principles, mechanistic pathways, and material systems. The article contrasts thermal ALE with ALD and plasma-based ALE, and develops mechanistic classifications such as fluorination–ligand exchange, conversion–etch, and oxidation–fluorination cycles. It also surveys materials and application challenges, including temperature-dependent EPC behavior, precursor design, and process selectivity. Within this broader context, Table 3 provides a cross-material summary of experimentally verified thermal ALE processes, which we use as the basis for a machine-actionable ORKG comparison.

3.6.1 Machine-actionable modeling of Table 3

Table 3 in Fang *et al.* (2018) [19] summarizes representative thermal ALE processes across multiple material classes. For each process, it reports the etched material, surface-adsorption reactant, surface-

removal reactant, etch-per-cycle (EPC) rate, and process temperature. We modeled this table in the ORKG as Comparison <https://orkg.org/comparisons/R1560222> [10], with each process represented as a separate `compareContribution` and the tabular fields encoded as comparison predicates. This machine-actionable representation provides the basis for the Easy and Complex Queries discussed below.

Table 21: Answer table for Q.24 over Table 3 of Fang et al. [19], modeled in the ORKG as Comparison R1560222. The table lists all thermal ALE processes with $EPC > 0.5 \text{ \AA/cycle}$, including the material, surface-adsorption reactant, surface-removal reactant, EPC, and etching temperature.

Material	Surface adsorption	Surface removal	EPC [$\text{\AA/cycle}$]	Temp [$^{\circ}\text{C}$]
W	O ₂	WF ₆	6.30	300
WO ₃	BCl ₃	HF	4.19	207
W	O ₃ /BCl ₃	HF	2.50	207
ZnO	HF	TMA	2.19	295
AlN	HF	Sn(acac) ₂ /H ₂ plasma	1.96	250
HfO ₂	HF	Al(CH ₃) ₂ Cl	0.77	250
Al ₂ O ₃	HF	Sn(acac) ₂	0.61	250
TiO ₂	WF ₆	BCl ₃	0.60	170

Easy Query. The 24th question is: “List all thermal ALE processes with $EPC > 0.5 \text{ \AA/cycle}$; return the material, surface-adsorption and surface-removal reactants, EPC, and etching temperature; sort by EPC.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t3-easy-fang>. The resulting high-EPC processes from the ORKG comparison are shown in Table 21.

Table 22: Answer table for Q.25 over Table 3 of Fang et al. [19], modeled in the ORKG as Comparison R1560222. For each mechanistic archetype, the table reports the number of distinct materials and the mean EPC (in $\text{\AA/cycle}$).

Mechanism archetype	# Materials	Mean EPC [$\text{\AA/cycle}$]
Other	2	5.245
Oxidation + fluorination	2	1.350
Fluorination + ligand-exchange	7	0.617
Halogenation + conversion	1	0.600

Complex Query. The 25th question is: “Group all thermal ALE processes by inferred mechanism archetype—(i) Fluorination + ligand-exchange (HF with TMA/DMAC/Sn(acac)₂/Al(CH₃)₂Cl/SiCl₄), (ii) Halogenation + conversion (WF₆ with BCl₃), (iii) Oxidation + fluorination (O₂/O₃ with HF), or (iv) Other—and compute, for each group, the number of distinct materials and the mean EPC.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t3-complex-fang>. The aggregated results are shown in Table 22.

3.7 Paper – Physical and chemical effects in directional atomic layer etching

Sang and Chang (2020) [56] review the physical and chemical factors that govern directional ALE, with emphasis on how ion-surface interactions, energy thresholds, sputtering, surface oxidation, ligand exchange, and reaction thermodynamics interact to produce anisotropy and selectivity. The article discusses plasma-based, thermal, and hybrid plasma-thermal ALE routes, linking fundamental ion-driven phenomena and surface reactions to integration requirements in advanced semiconductor patterning.

3.7.1 Machine-actionable modeling of Table 1

Table 1 in Sang and Chang (2020) [56] summarizes representative ALE processes for elemental and compound semiconductors, including Si, Ge, GaAs, InP, GaN, InGaAs, and AlGaIn. For each process, it records the etched material, activation mechanism (typically plasma-driven), surface-modification chemistry, removal step, and key process conditions. The table also includes two purely

thermal ALE systems, GaN ($\text{XeF}_2/\text{BCl}_3$) and InGaAs (HF -pyridine/ $\text{AlCl}(\text{CH}_3)_2$). We modeled this table in the ORKG as Comparison <https://orkg.org/comparisons/R1560825> [63]. This machine-actionable representation underlies the Easy and Complex Queries discussed below.

Table 23: Answer table for Q.26 over Table 1 of Sang and Chang [56], modeled in the ORKG as Comparison R1560825. The table lists semiconductor ALE processes with their modification, removal, and activation modes.

Semiconductor	Modification	Removal	Activation
AlGaIn	Cl_2	Ar	plasma
GaAs	Cl_2	Ar	plasma
GaN	Cl_2	Ar	plasma
Ge	Cl_2	Ar	plasma
InGaAs	Cl_2	Ar	plasma
InP	Cl_2	Ne	plasma
Si	Cl_2	Ar	plasma
GaN	XeF_2	BCl_3	Thermal
InGaAs	HF -pyridine	$\text{AlCl}(\text{CH}_3)_2$	Thermal

Easy Query. The 26th question is: “List all semiconductor ALE processes and return, for each, the modification, removal, and activation types; sort by activation mode.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t1-semi-ale>. The resulting semiconductor ALE processes are shown in Table 23.

Table 24: Answer table for Q.27 over Table 1 of Sang and Chang [56], modeled in the ORKG as Comparison R1560825. For each activation type, the table lists modification–removal pairs and the number of distinct semiconductor materials using each pair.

Activation	Modification / Removal pair	# Semiconductors
plasma	Cl_2 / Ar	6
plasma	Cl_2 / Ne	1
Thermal	HF -pyridine / $\text{AlCl}(\text{CH}_3)_2$	1
Thermal	XeF_2 / BCl_3	1

Complex Query. The 27th question is: “Group semiconductor ALE processes by activation type (plasma vs thermal) and count the number of distinct materials using each modification–removal pair.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t1-sang-complex>. The aggregated results are shown in Table 24.

3.8 Paper – Anisotropic/Isotropic Atomic Layer Etching of Metals

Kim *et al.* (2020) [55] review recent progress in ALE of metals, contrasting anisotropic, ion-assisted pathways with purely thermal, isotropic routes. Motivated by scaling challenges in advanced semiconductor devices and 3D architectures such as GAA and vertical FETs, the article positions ALE as a low-damage, angstrom-level alternative to conventional reactive ion etching. It surveys reported ALE chemistries for metals and metal nitrides (e.g., Cu, W, Co, Cr, Fe, Pt, TiN, AlN), discussing adsorption steps, ion-energy windows, thermal reactions, and process ideality. Within this context, Table 2 summarizes directional behavior, reactant pairings, process temperatures, and EPC values for metal ALE systems, and serves as the basis for our machine-actionable ORKG comparison.

3.8.1 Machine-actionable modeling of Table 2

Table 2 in Kim *et al.* (2020) [55] summarizes reported ALE processes for metals and metal nitrides. For each entry, the table lists the etched material, directionality (anisotropic or isotropic), surface-modification and removal reactants, process temperature, and EPC. Materials covered include Co, Fe, Pd, Pt, Cr, W, Cu, and nitrides such as TiN and AlN. We modeled this table in the ORKG as Comparison <https://orkg.org/comparisons/R1563131> [12], forming a machine-actionable dataset that supports the Easy and Complex Queries presented below.